5.3.1 User Interface / Input Handling Module

This module serves as the main entry point between the user and the system. It allows users to submit text data or documents, including PDF files, for analysis and validates that submitted input can be processed before it is passed further into the pipeline. Invalid or unsupported input is rejected or flagged at this stage, ensuring that only well-formed data reaches the detection components.

5.3.2 Rule-Based Filtering Module

This module applies configurable, extensible regular-expression patterns to detect known, structured identifiers such as email addresses, phone numbers and financial or credential-related patterns. It operates quickly and with high precision on the exact patterns it is configured to recognise, providing a fast first line of detection that complements the contextual analysis performed by the later modules.

5.3.3 NLP-Based Contextual Analysis Module

This module uses a fine-tuned, transformer-based model (BERT) to analyse the semantic context of the submitted input. Rather than relying on surface-level pattern matching, it evaluates the meaning of the surrounding text to distinguish genuinely sensitive content from superficially similar but harmless content, allowing it to catch unstructured or paraphrased leakage that the rule-based filter would miss.

5.3.4 Deep Learning-Based Pattern Detection Module

This module applies LSTM and/or CNN-based deep learning models to identify complex or hidden leakage patterns that are not explicitly defined by the rule-based filter and may not be fully captured by the contextual NLP model alone. It is particularly useful for recognising previously unseen or evolving leakage patterns based on learned structural and sequential characteristics of the input text.

5.3.5 Anomaly Detection Module

This module monitors data-handling behaviour, such as access frequency, transmission volume or timing, for deviations from established normal patterns. It supports the system's insider-threat detection capability by flagging behavioural anomalies as potential leakage incidents, independent of whether the transmitted content itself matches a known sensitive pattern.

5.3.6 Risk Scoring Module

This module combines the outputs of the rule-based, NLP, deep learning and anomaly detection components into a single, unified risk score. The score reflects the type, sensitivity and confidence of the detections made across all components, and is used to classify each analysed input and to determine whether an alert should be raised.

5.3.7 Explainable AI Module

This module uses explainability techniques such as SHAP and LIME to generate a human-readable rationale for every detection decision. By presenting the reasoning